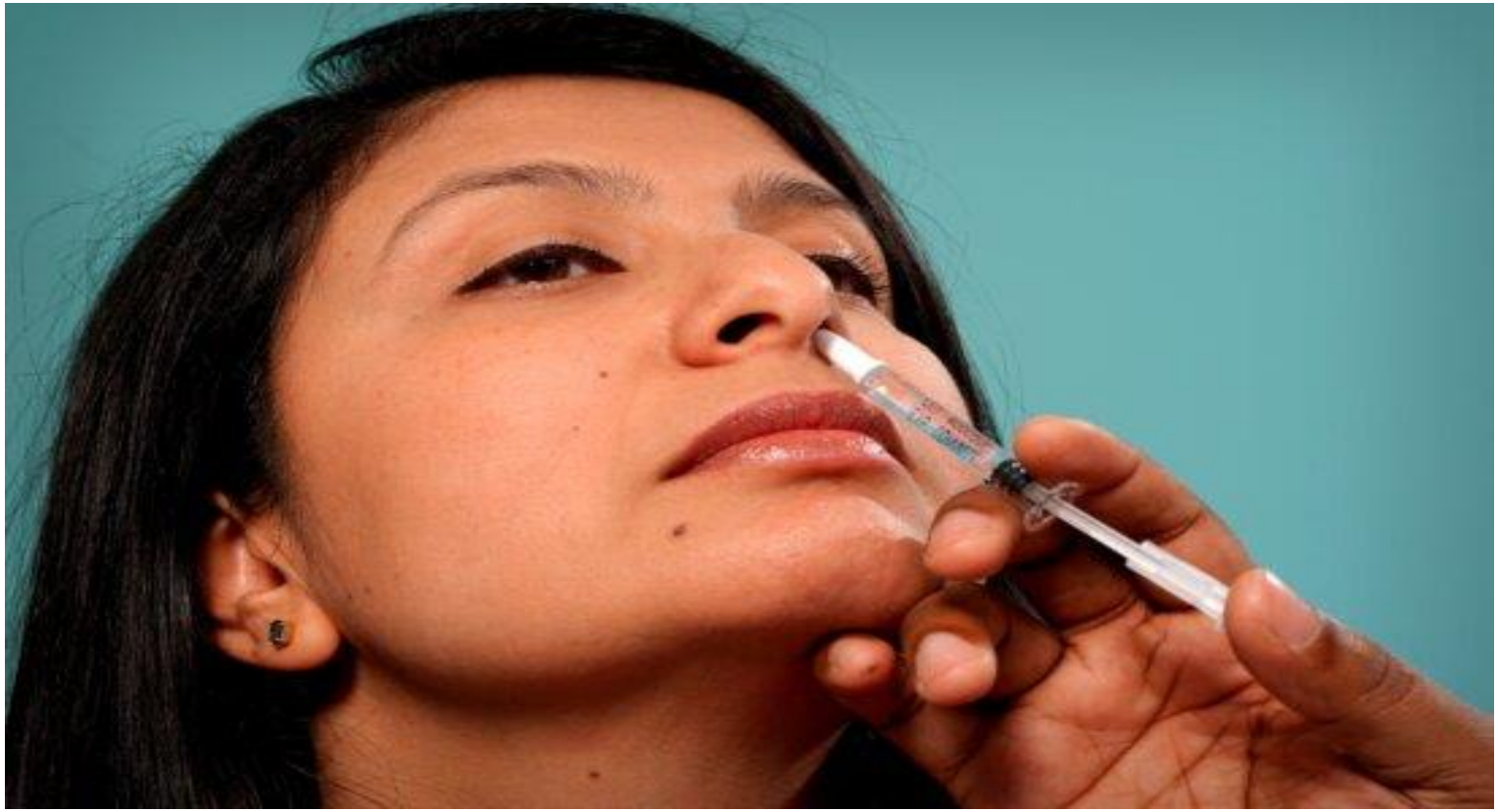
Predicting H1N1 Vaccine Uptake using Machine Learning



Business Problem

- Vaccine reluctance reduces immunization rates and increases the risk of infectious disease outbreaks such as H1N1.
- Identifying the key drivers behind individuals' vaccination decisions is essential for effective public health interventions.
- Public health organizations face challenges in effectively targeting vaccination efforts due to varying behaviors like demographics, and perceptions across populations.

Project Objectives

- Build a machine learning model that predicts whether an individual received the H1N1 vaccine.
- Identify the main factors that influence vaccine acceptance.
- Use the results from the predictive models to generate insights that can help public health professionals and policymakers design strategies to improve vaccination uptake.
- Evaluate the performance of different machine learning models by comparing how well they predict vaccine uptake.

Dataset Overview

- Source: National H1N1 Flu Survey

Features include:

- Demographics (age, race, education)
- Health conditions
- Behavioral practices
- Opinions about vaccines
- Doctor recommendations

- Target Variable: Whether an individual received the H1N1 vaccine.

Data Preparation

Key preprocessing steps:

- Handle missing values using imputation
- Encode categorical variables using One-Hot Encoding
- Scale numeric variables using StandardScaler
- Use pipelines to avoid data leakage
- Split dataset into training and validation sets

Modeling Approach

Two models were trained and compared:

- **Logistic Regression (Baseline Model)**

Built as the initial benchmark model

Provides interpretable insights into feature influence

Evaluated using accuracy, ROC-AUC, confusion matrix

- **Random Forest (Advanced Model)**

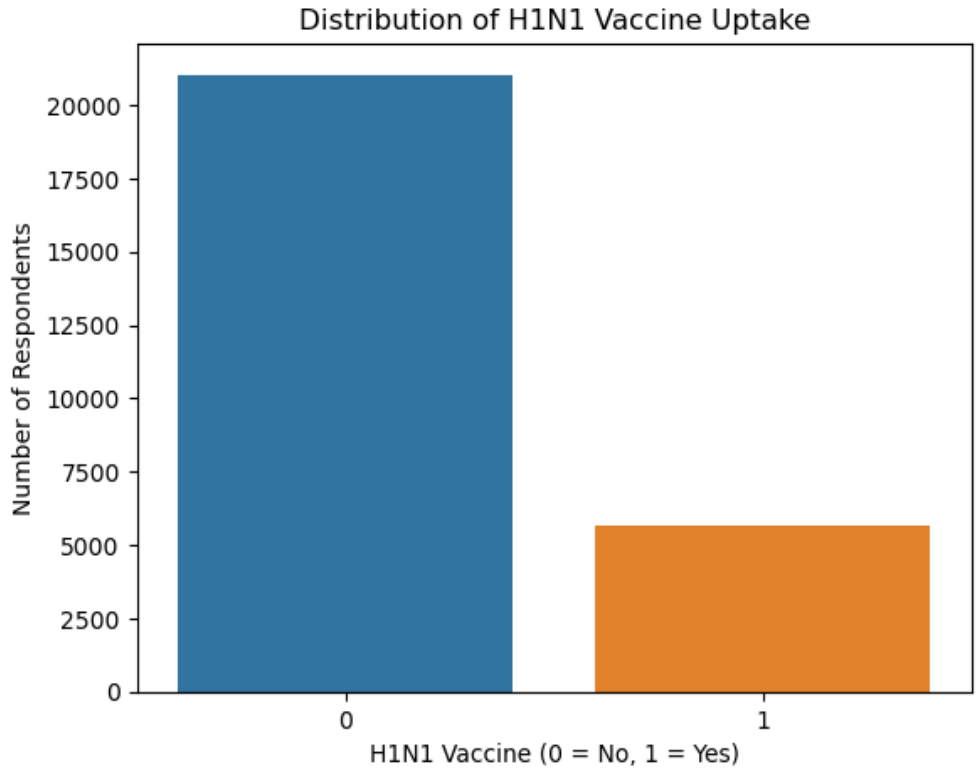
Ensemble learning method for improved performance

Captures complex and nonlinear relationships in the data

Compared against baseline using the same evaluation metrics

Model Evaluation

Models were evaluated u



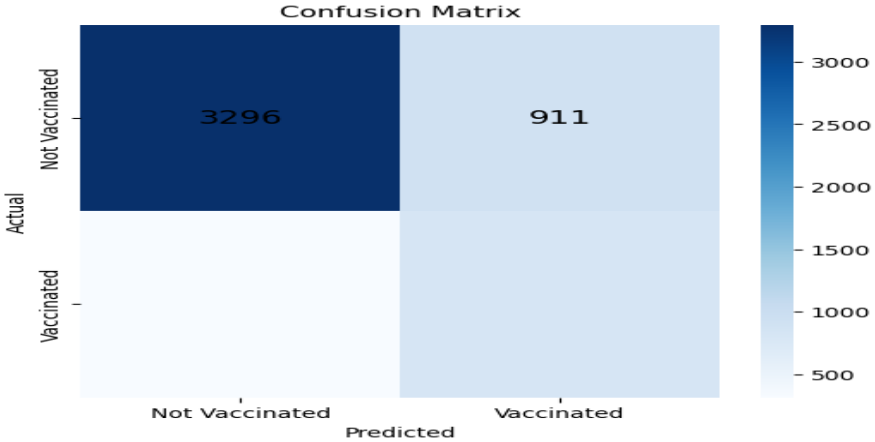
- • Accuracy
- • Precision
- • Recall
- • F1 Score
- • ROC-AUC

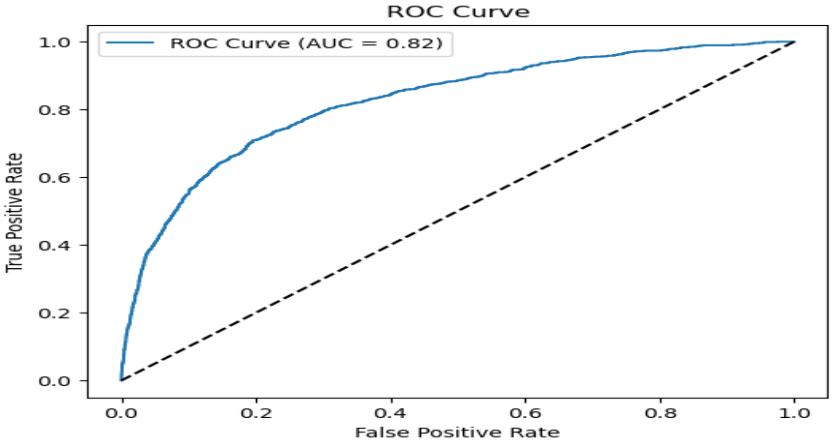
Since the dataset is imbalanced, ROC-AUC and F1 Score were emphasized.

Model Comparison

- The Random Forest model demonstrated improved performance in capturing complex relationships between behavioral and demographic factors.

- Logistic Regression provided a strong interpretable baseline.

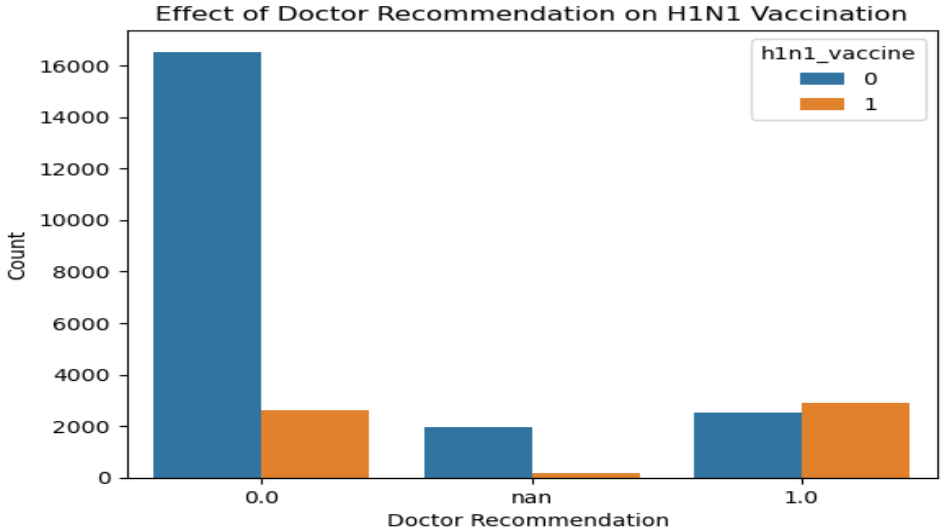




Key Insights

Important predictors of vaccination include:

- Doctor recommendation
- Perceived vaccine effectiveness
- Individual knowledge about H1N1



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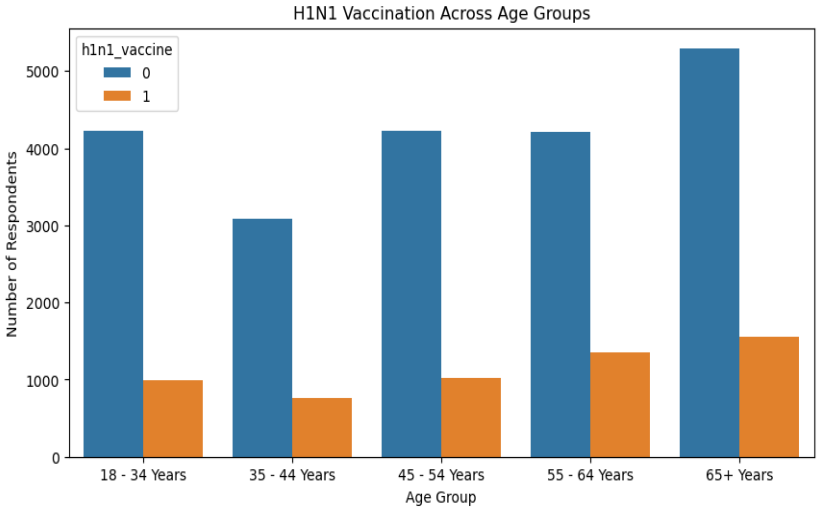

Recommendations

Based on the analysis:

- Strengthen doctor-led vaccine recommendations
- Improve public awareness campaigns on vaccine effectiveness
- Target outreach toward populations with lower predicted uptake

Future Improvements

Potential improvements include:

- Additional feature engineering
- Testing advanced models such as Gradient Boosting

Conclusions

- Addressing public health concerns and improving awareness is essential to increase vaccine uptake and encourage informed decision-making.
- Insights from the analysis can help public health authorities design targeted communication strategies